Drude again: Summary

$$\frac{d}{dt}\vec{v}_d = \frac{q}{m}\vec{E}_0 e^{-i\omega t} - \frac{\vec{v}_d}{\tau}$$

$$\vec{v}_d(t) = \vec{v}_d\, e^{-i\omega t}$$

$$-i\omega\,\vec{v}_d + \frac{1}{\tau}\vec{v}_d = \frac{q\vec{E}_0}{m}\cancel{e^{-i\omega t}}$$

$$\vec{v}_d = \frac{q\vec{E}_0}{m\left(-i\omega + \frac{1}{\tau}\right)}$$

$$\vec{v}_d(t) = \frac{q\vec{E}_0 e^{-i\omega t}}{m\left(-i\omega + \frac{1}{\tau}\right)}$$

$$\vec{j}(t) = qn\vec{v}_d = \frac{q^2 n\,\vec{E}_0 e^{-i\omega t}}{m\,\frac{1}{\tau}\left(1 - i\omega\tau\right)} = \sigma(\omega)\,\vec{E}_0 e^{-i\omega t}$$

Drude conductivity $\sigma(\omega) = \dfrac{e^2 n\tau}{m(1-i\omega\tau)} = \dfrac{\sigma_0}{1-i\omega\tau}$

Dielectric function $\varepsilon(\omega) = \varepsilon_0 + \dfrac{i\sigma(\omega)}{\omega}$

$$\varepsilon(\omega) = \varepsilon_0 + \frac{i q^2 n}{m\omega\left(-i\omega + \frac{1}{\tau}\right)} = \varepsilon_0 - \frac{i q^2 n}{i m\omega\left(\omega + \frac{i}{\tau}\right)}$$

$$= \varepsilon_0\left(1 - \frac{q^2 n}{\varepsilon_0 m}\,\frac{1}{\omega\left(\omega + \frac{i}{\tau}\right)}\right)$$

So that $\varepsilon(\omega) = \varepsilon_0\left(1 - \dfrac{\omega_p^2}{\omega\left(\omega + \frac{i}{\tau}\right)}\right)$, $\omega_p = \sqrt{\dfrac{e^2 n}{\varepsilon_0 m}}$

in conductors

plasma frequency

In dielectrics: $\varepsilon(\omega) = \varepsilon_0\left(1 + \dfrac{\omega_p^2}{\omega_0^2 - \omega^2 - i\frac{\omega}{\tau}}\right)$

where ω_0 = restoring frequency ($\omega_0 = 0$ in conductors)

Maxwell's eqns:

(1) $\vec{\nabla}\times\vec{E} + \frac{\partial}{\partial t}\vec{B} = 0 \Rightarrow i\vec{k}\times\vec{E}(\omega,\vec{k}) - i\omega\,\vec{B}(\omega,\vec{k}) = 0$

$\vec{k}\times\vec{E} = \omega\vec{B}$ for Fourier transforms of $\vec{E}(r,t)$ and $\vec{B}(r,t)$.

Similarly

$\vec{\nabla}\cdot\vec{B} = 0 \Rightarrow \vec{k}\cdot\vec{B} = 0$

$\vec{\nabla}\times\vec{B} = \mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t} + \mu_0\vec{J} \Rightarrow i\vec{k}\times\vec{B} = -\mu_0\varepsilon_0 i\omega\vec{E} + \mu_0(\vec{j}_{ext} + \vec{j}_{ind})$

$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0} = \frac{\rho_{ext}+\rho_{ind}}{\varepsilon_0} \Rightarrow i\vec{k}\cdot\vec{E} = \frac{1}{\varepsilon_0}(\rho_{ext}+\rho_{ind})$

Using $\rho_{ind} = -\vec{\nabla}\cdot\vec{P} \Rightarrow \rho_{ind}(\omega,k) = -i\vec{k}\cdot\vec{P}(\omega,\vec{k})$

$\vec{j}_{ind} = \frac{\partial\vec{P}}{\partial t} + \vec{\nabla}\times\vec{M} \Rightarrow \vec{j}_{ind} = -i\omega\vec{P} + i\vec{k}\times\vec{M}$

We obtain in Fourier space (for linear medium)

$$\begin{cases} \vec{k}\times\vec{E} = \omega\vec{B} \\ \vec{k}\cdot\vec{B} = 0 \\ i\vec{k}\times\vec{H} = i\frac{1}{\mu(\omega,k)}\vec{k}\times\vec{B} = -i\omega\vec{D} + \vec{j}_{ext} & \text{(I)} \\ \qquad = -i\omega\varepsilon\vec{E} + \vec{j}_{ext} & (3) \\ i\vec{k}\cdot\vec{D} = i\,\vec{k}\cdot\vec{E}\,\varepsilon = \rho_{ext} & (4) \end{cases}$$

$\vec{M} = \chi_m\vec{H} \Rightarrow \vec{B} = \mu_0(1+\chi_m)\vec{H} = \mu\vec{H}$

While in terms of $\vec{\mathcal{D}}$:

$$\begin{cases} \vec{k} \times \vec{E} = \omega \vec{B} \\ \vec{k} \cdot \vec{B} = 0 & (II) \\ i\, \vec{k} \times \vec{B} = -i\omega\mu_0 \vec{\mathcal{D}} + \mu_0 \vec{j}_{ext} & (3) \\ i\, \vec{k} \cdot \vec{\mathcal{D}} = i\, \vec{k} \cdot \vec{E}\, \tilde{\varepsilon} = \rho_{ext} & (4) \end{cases}$$

To connect two descriptions we need to understand structure of $\tilde{\varepsilon}(\omega, \vec{k})$: can form tensor even in isotropic medium

generally $\tilde{\varepsilon}_{\alpha\beta}(\omega, \vec{k}) = A_1(\omega, \vec{k})\, \delta_{\alpha\beta} + A_2(\omega, \vec{k})\, \dfrac{k_\alpha k_\beta}{k^2}$

where $A_{1,2}(\omega, \vec{k}) = A_{1,2}(\omega, k)$

$\hookrightarrow k = |\vec{k}|$

$$\mathcal{D}_\alpha = \tilde{\varepsilon}_{\alpha\beta} E_\beta \Rightarrow \mathcal{D}_\alpha = A_1 E_\alpha + A_2 \frac{k_\alpha (\vec{k} \cdot \vec{E})}{k^2}$$

$$\vec{\mathcal{D}} = A_1 \vec{E} + A_2 \frac{\vec{k}(\vec{k} \cdot \vec{E})}{k^2}$$

That is, response for <u>transverse</u> and <u>longitudinal</u> fields are different!

For $\vec{k} \cdot \vec{E} = 0$, so that $\vec{E} \perp \vec{k}$, (transverse)

$$\vec{\mathcal{D}} = A_1 \vec{E} = \tilde{\varepsilon}^{tr}(\omega, \vec{k})\, \vec{E}$$

For $\vec{E} \parallel \vec{k}$, $\vec{\mathcal{D}} = (A_1 + A_2)\, \vec{E} = \tilde{\varepsilon}^{l}(\omega, k)\, \vec{E}$

longitudinal dielectric function

$$A_2 = \tilde{\varepsilon}^{\ell} - A_1 = \tilde{\varepsilon}^{\ell} - \tilde{\varepsilon}^{tr}$$

$$\Rightarrow \underset{(\omega,k)}{\tilde{\varepsilon}} = \tilde{\varepsilon}^{tr}\,\delta_{\alpha\beta} + \left(\tilde{\varepsilon}^{\ell} - \tilde{\varepsilon}^{tr}\right)\frac{k_\alpha k_\beta}{k^2}$$

$$= \underset{(\omega,k)}{\tilde{\varepsilon}^{tr}}\left(\delta_{\alpha\beta} - \frac{k_\alpha k_\beta}{k^2}\right) + \underset{(\omega,k)}{\tilde{\varepsilon}^{\ell}}\,\frac{k_\alpha k_\beta}{k^2}$$

Compare eqns (4) in systems I and II:

$$i\,\vec{k}\cdot\left(\vec{E}\,\varepsilon\right) = \rho_{ext}$$

$$i\,k_a D_a = \rho_{ext}$$

$$D_a = \varepsilon_{ab}\,E_b$$

and $\quad i\,\vec{k}\cdot\left(\vec{E}\,\tilde{\varepsilon}\right) = \rho_{ext}$

$$i\,\vec{k}\cdot\vec{\mathcal{D}} = \rho_{ext}$$

In components: $\quad i\,k_a \mathcal{D}_a = i\,k_a\left(\tilde{\varepsilon}_{ab}\,E_b\right) =$

$$= i\,k_a\left\{\left(\delta_{ab} - \frac{k_a k_b}{k^2}\right)\tilde{\varepsilon}^{tr}\right.$$

$$\left. + \tilde{\varepsilon}_{\ell}\,\frac{k_a k_b}{k^2}\right\} E_b = i\,k_b E_b\,\tilde{\varepsilon}_{\ell} = \rho_{ext}$$

but $\quad k_a\left(\delta_{ab} - \dfrac{k_a k_b}{k^2}\right) = k_b - \dfrac{k_a k_a k_b}{k^2} =$

$$= k_b - k_b = 0$$

while $\quad k_a\,\dfrac{k_a k_b}{k^2} = \dfrac{k_a k_a}{k^2}\,k_b = k_b$

Therefore $\varepsilon(\omega,k) = \tilde{\varepsilon}_\ell(\omega,k)$ (longitudinal)

Now look at eq (3):

I $\quad \frac{i}{\mu}\vec{k}\times\vec{B} = -i\omega\,\varepsilon\,\vec{E} + \vec{j}_{ext} \Rightarrow i\,\frac{\mu_0}{\mu}\,\vec{k}\times\vec{B} = -i\mu_0\omega\varepsilon\vec{E} + \mu_0\vec{j}_{ext}$

II $\quad i\vec{k}\times\vec{B} = -i\omega\mu_0\underbrace{\tilde{\varepsilon}\,\vec{E}}_{\vec{D}} + \mu_0\vec{j}_{ext}$

$$\vec{D} = \tilde{\varepsilon}_{tr}\left(\vec{E} - \frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2}\right) + \tilde{\varepsilon}_\ell\,\frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2}$$

Subtract(μ_0 I) from II:

$$i\left(1-\frac{\mu_0}{\mu}\right)\vec{k}\times\vec{B} = -i\omega\mu_0\left[\tilde{\varepsilon}_{tr}\left(\vec{E}-\frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2}\right) + \tilde{\varepsilon}_\ell\,\frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2} - \varepsilon\vec{E}\right]$$

but $\varepsilon = \tilde{\varepsilon}_\ell \Rightarrow$

$$i\left(1-\frac{\mu_0}{\mu}\right)\vec{k}\times\vec{B} = -i\omega\mu_0\left(\tilde{\varepsilon}_{tr}-\tilde{\varepsilon}_\ell\right)\left(\vec{E}-\frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2}\right)$$

From Faraday's law: $\vec{B} = \frac{1}{\omega}\vec{k}\times\vec{E} \Rightarrow \vec{k}\times\vec{B} = \frac{1}{\omega}\vec{k}\times(\vec{k}\times\vec{E})$

$$= \frac{1}{\omega}\left\{\vec{k}(\vec{k}\cdot\vec{E}) - k^2\vec{E}\right\} =$$

$$= -\frac{k^2}{\omega}\left(\vec{E} - \frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2}\right)$$

Therefore

$$\cancel{i}\left(1-\frac{\mu_0}{\mu}\right)\frac{k^2}{\omega} = \cancel{i}\omega\mu_0\left(\tilde{\varepsilon}_{tr}-\tilde{\varepsilon}_l\right)$$

$$1-\frac{\mu_0}{\mu(k,\omega)} = \frac{\omega^2}{k^2}\mu_0\left(\tilde{\varepsilon}_{tr}-\tilde{\varepsilon}_l\right) = \frac{\omega^2}{c^2k^2}\left(\frac{\tilde{\varepsilon}_{tr}-\tilde{\varepsilon}_l}{\varepsilon_0}\right)$$

$$\frac{\mu(k,\omega)}{\mu_0} = \left\{1-\frac{\omega^2}{c^2k^2}\left[\frac{\tilde{\varepsilon}_{tr}(k,\omega)-\tilde{\varepsilon}_l(k,\omega)}{\varepsilon_0}\right]\right\}^{-1}$$

We observe that difference between transverse $\tilde{\varepsilon}_{tr}(k,\omega)$ and longitudinal $\tilde{\varepsilon}_l(k,\omega)$ dielectric functions is due to magnetic permiability $\mu(k,\omega)=\mu_0\left(1+\chi_m(k,\omega)\right)$.

If $\mu(k,\omega)=\mu_0$, then there is no difference, $\tilde{\varepsilon}_{tr}=\tilde{\varepsilon}_l=\varepsilon(k,\omega)$.

Therefore ~~any person~~ every material is characterized by two material-dependent quantities:

1) at $\omega=0$ (static limit) we need $\varepsilon=\varepsilon_0(1+\chi_e)$
and $\mu=\mu_0(1+\chi_m)$

2) at finite $\omega\neq 0$, we need $\varepsilon(k,\omega)$ and $\mu(k,\omega)$

OR

we can proceed without $\vec{B}$ at all, since $\omega\vec{B}=\vec{k}\times\vec{E}$. Then we need $\tilde{\varepsilon}_{tr}(k,\omega)$ and $\tilde{\varepsilon}_l(k,\omega)$.

Can also express $\tilde{\varepsilon}$ in terms of ε, μ:

$$\tilde{\varepsilon}_{\alpha\beta} = \left(\delta_{\alpha\beta} - \frac{k_\alpha k_\beta}{k^2}\right) \tilde{\varepsilon}_{tr} + \frac{k_\alpha k_\beta}{k^2} \underset{\varepsilon}{\underset{\parallel}{\tilde{\varepsilon}_\ell}}$$

$$\frac{\tilde{\varepsilon}_{tr}}{\varepsilon_0} = \frac{c^2 k^2}{\omega^2}\left(1 - \frac{\mu_0}{\mu}\right) + \frac{\tilde{\varepsilon}_\ell}{\varepsilon_0}$$

$$\Rightarrow \tilde{\varepsilon}_{tr} = \frac{c^2 k^2}{\omega^2} \varepsilon_0 \left(1 - \frac{\mu_0}{\mu}\right) + \tilde{\varepsilon} = \frac{k^2}{\omega^2}\left(\frac{1}{\mu_0} - \frac{1}{\mu}\right) + \tilde{\varepsilon}$$

$$c^2 \varepsilon_0 = \frac{\varepsilon_0}{\varepsilon_0 \mu_0} = \frac{1}{\mu_0}$$

$$\Rightarrow \tilde{\varepsilon}_{\alpha\beta} = \delta_{\alpha\beta}\, \varepsilon + \frac{k^2}{\omega^2}\left(\frac{1}{\mu_0} - \frac{1}{\mu}\right)\left(\delta_{\alpha\beta} - \frac{k_\alpha k_\beta}{k^2}\right).$$